

Calculus and Linear Algebra

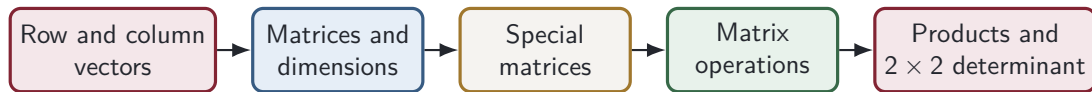
Matrices and Matrix Operations

Lecture 4

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Lecture outline



Aim

To introduce matrices as rectangular arrays of numbers, define their basic operations, and connect the determinant of a 2×2 matrix with geometry.

Row vectors and column vectors

A vector with n components can be written in two forms.

row vector $\mathbf{x}^\top = (x_1 \ x_2 \ \cdots \ x_n),$

column vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$

The symbol $^\top$ represents the operation of *transposition*. $\mathbf{x}^\top$ is said to be *transpose* of the vector $\mathbf{x}$.

row vector

$$1 \times n$$

column vector

$$n \times 1$$

The transpose changes orientation

For

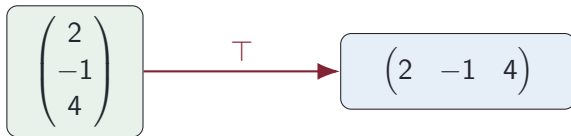
$$\mathbf{x} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix},$$

the transpose is

$$\mathbf{x}^{\top} = (2 \quad -1 \quad 4).$$

In general,

$$(\mathbf{x}^{\top})^{\top} = \mathbf{x}.$$



Exercise: row or column?

Exercise

For each vector, state whether it is a row vector or a column vector and give its dimensions.

$$\mathbf{u} = (3 \quad 0 \quad -2 \quad 5), \quad \mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ -1 \end{pmatrix}, \quad \mathbf{w} = (7).$$

Compute $\mathbf{u}^\top$ and $\mathbf{v}^\top$.

Exercise: solution

Solution

$$\mathbf{u} : 1 \times 4, \quad \mathbf{v} : 3 \times 1, \quad \mathbf{w} : 1 \times 1.$$

$$\mathbf{u}^\top = \begin{pmatrix} 3 \\ 0 \\ -2 \\ 5 \end{pmatrix}, \quad \mathbf{v}^\top = \begin{pmatrix} 1 & 4 & -1 \end{pmatrix}.$$

Matrices

Definition

An $m \times n$ **matrix** is a rectangular array of numbers with m rows and n columns:

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The entry a_{ij} lies in row i and column j .

Dimensions of a matrix

Let

$$A = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{pmatrix}.$$

Then A has

2 rows, 3 columns,

so

$$A \in \mathbb{R}^{2 \times 3}.$$

$$\begin{array}{c} \text{3 columns} \\ \overbrace{\hspace{10em}} \\ \text{2 rows} \left\{ \begin{array}{c} \boxed{\begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{pmatrix}} \end{array} \right. \end{array}$$

Rows, columns and entries

For

$$A = \begin{pmatrix} 1 & 4 & -2 \\ 3 & 0 & 5 \\ 7 & -1 & 6 \end{pmatrix},$$

we have

$$a_{23} = 5, \quad \text{row}_2(A) = (3 \ 0 \ 5), \quad \text{col}_1(A) = \begin{pmatrix} 1 \\ 3 \\ 7 \end{pmatrix}.$$

First index = row, second index = column.

Exercise: matrix notation

Exercise

Consider

$$B = \begin{pmatrix} 2 & 0 & -3 & 1 \\ 5 & 4 & 2 & -1 \\ 0 & 7 & 6 & 3 \end{pmatrix}.$$

Determine:

1. the dimensions of B ;
2. b_{12} , b_{24} , b_{32} ;
3. the second row;
4. the third column.

Exercise: solution

Solution

$$B \in \mathbb{R}^{3 \times 4}.$$

$$b_{12} = 0, \quad b_{24} = -1, \quad b_{32} = 7.$$

$$\text{row}_2(B) = (5 \quad 4 \quad 2 \quad -1), \quad \text{col}_3(B) = \begin{pmatrix} -3 \\ 2 \\ 6 \end{pmatrix}.$$

Rectangular and square matrices

A matrix is **square** if it has the same number of rows and columns.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in \mathbb{R}^{2 \times 2}, \quad \begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & 4 \\ 0 & 2 & 5 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

A matrix with $m \neq n$ is **rectangular**.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3}.$$

Remark: Only square matrices have a determinant.

The zero matrix

The **zero matrix** has every entry equal to zero.

$$0_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad 0_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

For every $A \in \mathbb{R}^{m \times n}$,

$$A + 0_{m \times n} = A.$$

The identity matrix

The **identity matrix** I_n is square, with 1 on the main diagonal and 0 elsewhere.

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is the multiplicative identity:

$$AI_n = A, \quad I_mA = A$$

whenever the products are defined.

Main and secondary diagonals

For a square matrix

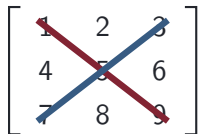
$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

the **main diagonal** is

$$a_{11}, a_{22}, a_{33},$$

and the **secondary diagonal** is

$$a_{13}, a_{22}, a_{31}.$$


$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

main diagonal

secondary diagonal

Diagonal and scalar matrices

A square matrix is **diagonal** if every off-diagonal entry is zero:

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

A **scalar matrix** is a diagonal matrix with the same value on the diagonal:

$$3I_3 = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Upper and lower triangular matrices

An **upper triangular matrix** has zeros below the main diagonal:

$$U = \begin{pmatrix} 2 & 1 & -3 \\ 0 & 4 & 5 \\ 0 & 0 & 7 \end{pmatrix}.$$

A **lower triangular matrix** has zeros above the main diagonal:

$$L = \begin{pmatrix} 2 & 0 & 0 \\ -1 & 4 & 0 \\ 3 & 5 & 7 \end{pmatrix}.$$

A diagonal matrix is both upper and lower triangular.

Symmetric matrices

A square matrix A is **symmetric** if

$$A^T = A.$$

Example:

$$A = \begin{pmatrix} 2 & -1 & 4 \\ -1 & 3 & 0 \\ 4 & 0 & 5 \end{pmatrix}.$$

The entries satisfy

$$a_{ij} = a_{ji}.$$

Hence a symmetric matrix is mirrored across the main diagonal.

Exercise: classify the matrices

Exercise

Classify each matrix using all applicable labels.

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 4 & 5 \\ 0 & 0 & 2 \end{pmatrix},$$
$$C = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & -1 \\ -1 & 5 \end{pmatrix}.$$

Possible labels: square, rectangular, zero, identity, diagonal, scalar, upper triangular, lower triangular, symmetric.

Exercise: solution

Solution

	square	zero	diagonal	scalar	triangular	symmetric
A	yes	no	yes	yes	both	yes
B	yes	no	no	no	upper	no
C	no	yes	–	–	–	–
D	yes	no	no	no	no	yes

Equality of matrices

Two matrices are equal if they have the same dimensions and all corresponding entries are equal.

$$A = B \iff A, B \in \mathbb{R}^{m \times n} \text{ and } a_{ij} = b_{ij} \forall i, j.$$

Example:

$$\begin{pmatrix} x & 2 \\ 3 & y \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & -4 \end{pmatrix} \implies x = 1, y = -4.$$

Matrix addition

Matrices can be added only if they have the **same dimensions**.

$$A + B = (a_{ij} + b_{ij}).$$

Example:

$$\begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} + \begin{pmatrix} 4 & 0 \\ 2 & -5 \end{pmatrix} = \begin{pmatrix} 5 & 2 \\ 1 & -2 \end{pmatrix}.$$

If $A \in \mathbb{R}^{2 \times 3}$ and $B \in \mathbb{R}^{3 \times 2}$, then $A + B$ is not defined.

Scalar multiplication

For $\lambda \in \mathbb{R}$ and $A = (a_{ij})$,

$$\lambda A = (\lambda a_{ij}).$$

Example:

$$-2 \begin{pmatrix} 1 & -3 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} -2 & 6 \\ 0 & -8 \end{pmatrix}.$$

Each entry is multiplied by the same scalar.

The transpose of a matrix

For $A \in \mathbb{R}^{m \times n}$, the transpose $A^T \in \mathbb{R}^{n \times m}$ is obtained by exchanging rows and columns.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \implies A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

In entries,

$$(A^T)_{ij} = a_{ji}.$$

Basic transpose identities

For matrices of compatible dimensions and $\lambda \in \mathbb{R}$,

$$(A^\top)^\top = A,$$

$$(A + B)^\top = A^\top + B^\top,$$

$$(\lambda A)^\top = \lambda A^\top.$$

For matrix products,

$$(AB)^\top = B^\top A^\top.$$

The order is reversed.

Exercise: matrix operations

Exercise

Let

$$A = \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 1 \\ -1 & 5 \end{pmatrix}.$$

Compute:

$$A + B, \quad A - B, \quad 3A, \quad A^{\top}, \quad (A + B)^{\top}.$$

Exercise: solution

Solution

$$A + B = \begin{pmatrix} 5 & -1 \\ 2 & 5 \end{pmatrix}, \quad A - B = \begin{pmatrix} -3 & -3 \\ 4 & -5 \end{pmatrix},$$

$$3A = \begin{pmatrix} 3 & -6 \\ 9 & 0 \end{pmatrix}, \quad A^{\top} = \begin{pmatrix} 1 & 3 \\ -2 & 0 \end{pmatrix},$$

$$(A + B)^{\top} = \begin{pmatrix} 5 & 2 \\ -1 & 5 \end{pmatrix}.$$

Matrix multiplication: compatibility

Let

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{n \times p}.$$

Then the product AB is defined and

$$AB \in \mathbb{R}^{m \times p}.$$



inner dimensions must agree

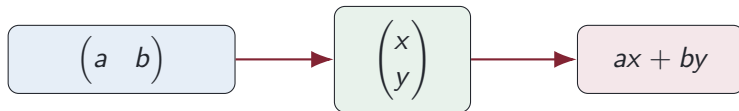
Row-by-column product

For

$$A = \begin{pmatrix} a & b \end{pmatrix}, \quad B = \begin{pmatrix} x \\ y \end{pmatrix},$$

we have

$$AB = ax + by.$$



Each entry of a matrix product is obtained by a row-by-column product.

2×2 matrix product

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}.$$

The entry $(AB)_{ij}$ is the product of row i of A with column j of B .

A numerical example

$$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 0 \\ 2 & 5 \end{pmatrix}.$$

Hence

$$AB = \begin{pmatrix} 1 \cdot 4 + 2 \cdot 2 & 1 \cdot 0 + 2 \cdot 5 \\ 3 \cdot 4 + (-1) \cdot 2 & 3 \cdot 0 + (-1) \cdot 5 \end{pmatrix} = \begin{pmatrix} 8 & 10 \\ 10 & -5 \end{pmatrix}.$$

A 2×3 by 3×2 product

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 0 & -2 \\ 3 & 5 \end{pmatrix}.$$

Dimensions:

$$(2 \times 3)(3 \times 2) \longrightarrow 2 \times 2.$$

For example,

$$(AB)_{12} = \begin{pmatrix} 1 & 2 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix} = 1 - 4 = -3.$$

Matrix multiplication is not commutative

In general,

$$AB \neq BA.$$

Example:

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Even when both products exist, they need not be equal.

Identity and multiplication

For every $A \in \mathbb{R}^{m \times n}$,

$$I_m A = A, \quad A I_n = A.$$

Example:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{pmatrix}.$$

The dimensions of the identity matrix depend on the side of multiplication.

Exercise: are the products defined?

Exercise

Suppose

$$A \in \mathbb{R}^{2 \times 3}, \quad B \in \mathbb{R}^{3 \times 4}, \quad C \in \mathbb{R}^{4 \times 2}.$$

For each product, decide whether it is defined. If it is, state the dimensions of the result.

$$AB, \quad BA, \quad BC, \quad CB, \quad AC, \quad CA.$$

Exercise: solution

Solution

$$AB : (2 \times 3)(3 \times 4) \rightarrow 2 \times 4,$$

$$BA : \text{not defined},$$

$$BC : (3 \times 4)(4 \times 2) \rightarrow 3 \times 2,$$

$$CB : \text{not defined},$$

$$AC : \text{not defined}, \quad CA : (4 \times 2)(2 \times 3) \rightarrow 4 \times 3.$$

Exercise: compute a matrix product

Exercise

Compute

$$AB,$$

where

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 2 \end{pmatrix}.$$

Then identify the row-by-column calculation that produces $(AB)_{21}$.

Exercise: solution

Solution

$$\begin{aligned} AB &= \begin{pmatrix} 1 \cdot 2 + 2(-1) + (-1)3 & 1 \cdot 1 + 2 \cdot 0 + (-1)2 \\ 0 \cdot 2 + 3(-1) + 4 \cdot 3 & 0 \cdot 1 + 3 \cdot 0 + 4 \cdot 2 \end{pmatrix} \\ &= \begin{pmatrix} -3 & -1 \\ 9 & 8 \end{pmatrix}. \end{aligned}$$

Moreover,

$$(AB)_{21} = (0 \quad 3 \quad 4) \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = 9.$$

The determinant of a 2×2 matrix

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the determinant is

$$\boxed{\det(A) = ad - bc}.$$

Equivalently,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

The determinant is a number associated with a square matrix.

Columns as vectors

Write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} | & | \\ \mathbf{u} & \mathbf{v} \\ | & | \end{pmatrix},$$

where

$$\mathbf{u} = \begin{pmatrix} a \\ c \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} b \\ d \end{pmatrix}.$$

The two columns determine a parallelogram in the plane.

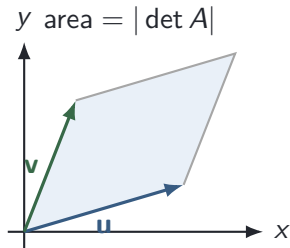
Determinant as signed area

For

$$A = \begin{pmatrix} | & | \\ \mathbf{u} & \mathbf{v} \\ | & | \end{pmatrix},$$

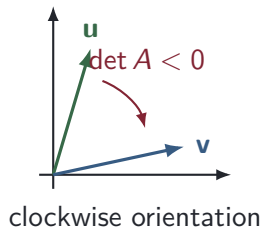
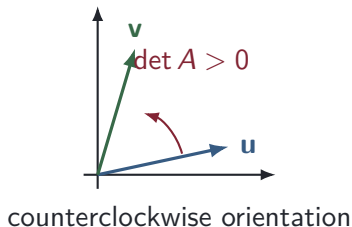
the columns generate a parallelogram. Its area is

$$|\det(A)|.$$



Orientation and the sign of the determinant

The sign records the orientation of the ordered pair of columns.



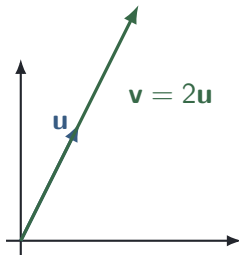
Zero determinant (!!!)

If the columns are linearly dependent, the parallelogram collapses to a segment.

$$\det(A) = 0.$$

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \quad \det(A) = 1 \cdot 4 - 2 \cdot 2 = 0.$$



Example: area and orientation

Let

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Then

$$\det(A) = 3 \cdot 2 - 1 \cdot 1 = 5.$$

Therefore:

- ▶ the parallelogram generated by the columns has area 5;
- ▶ the orientation is positive.

If the columns are exchanged, the determinant becomes -5 while the area remains 5.

Exercise: determinants

Exercise

Compute the determinant and interpret its sign.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix}.$$

For each matrix, state the area of the parallelogram generated by its columns.

Exercise: solution

Solution

$$\det(A) = 2 \cdot 3 - 1 \cdot 1 = 5, \quad \text{area} = 5,$$

$$\det(B) = 1 \cdot 1 - 3 \cdot 2 = -5, \quad \text{area} = 5,$$

$$\det(C) = 2(-2) - 4(-1) = 0, \quad \text{area} = 0.$$

Thus A has positive orientation, B negative orientation, and the columns of C are linearly dependent.

Integrated exercise

Exercise

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ -1 & 5 \end{pmatrix}.$$

Compute:

1. $A + B$ and $2A - B$;
2. A^T and B^T ;
3. AB and BA ;
4. $\det(A)$ and $\det(B)$;
5. the area of the parallelogram generated by the columns of each matrix.

Integrated exercise: solution

Solution

$$A + B = \begin{pmatrix} 3 & 2 \\ 2 & 9 \end{pmatrix}, \quad 2A - B = \begin{pmatrix} 0 & 4 \\ 7 & 3 \end{pmatrix}.$$

$$AB = \begin{pmatrix} 0 & 10 \\ 2 & 20 \end{pmatrix}, \quad BA = \begin{pmatrix} 2 & 4 \\ 14 & 18 \end{pmatrix}.$$

$$\det(A) = 1 \cdot 4 - 2 \cdot 3 = -2, \quad \det(B) = 2 \cdot 5 - 0(-1) = 10.$$

The corresponding areas are 2 and 10.

Summary

Matrices

- ▶ row and column vectors;
- ▶ dimensions $m \times n$;
- ▶ entries, rows and columns;
- ▶ square and special matrices;
- ▶ main and secondary diagonals.

Operations

- ▶ addition and scalar multiplication;
- ▶ transpose;
- ▶ row-by-column multiplication;
- ▶ 2×2 determinant;
- ▶ area and orientation.

Next: determinants, minors and the role of invertibility.